

Spendly

Offline-First Zero-Knowledge Peer-to-Peer Finance Ecosystem

Student Name · Enrollment Number · Semester VII · Information
Technology · College Name



The Problem We're Solving

Modern finance is built on fragile assumptions — always-online, always-trusted, always-exposed.

Privacy at Risk

Centralized servers store every transaction, creating permanent vulnerability to breaches and surveillance.

Cloud Dependency

Offline scenarios break billing, inventory, and payments entirely — a critical failure for real-world commerce.

High POS Costs

Monthly SaaS fees for point-of-sale systems are prohibitive for small merchants and independent retailers.





Introducing the Spendly Ecosystem

Three applications. One unified vision. Zero compromise on privacy or performance.

Spendly

Personal finance and peer-to-peer payments — entirely offline-first, zero-knowledge architecture, biometric-secured.

Spendly Shop

Merchant billing, inventory management, and QR invoice generation — no internet, no monthly fees.

Spendly Radar

Proximity-based offer discovery and coupon claiming through encrypted local peer-to-peer communication.

System Architecture



A layered, offline-first architecture where every component operates independently — and synchronizes only when trusted peers are present.

→ React + PWA

Responsive interface, installable on any device without app stores.

→ WebRTC + PeerJS

Direct device-to-device communication, no relay server required.

→ IndexedDB + Dexie.js

Encrypted local database — data never leaves the device without consent.

→ Cloudflare Workers + KV

Edge compute for lightweight sync — zero persistent server infrastructure.

How It Works

01

Discover Offers

Spendly Radar detects nearby merchant offers through encrypted local broadcast.

02

Claim Coupon · Scan Product

User claims offer; merchant scans product barcode or SKU for instant lookup.

03

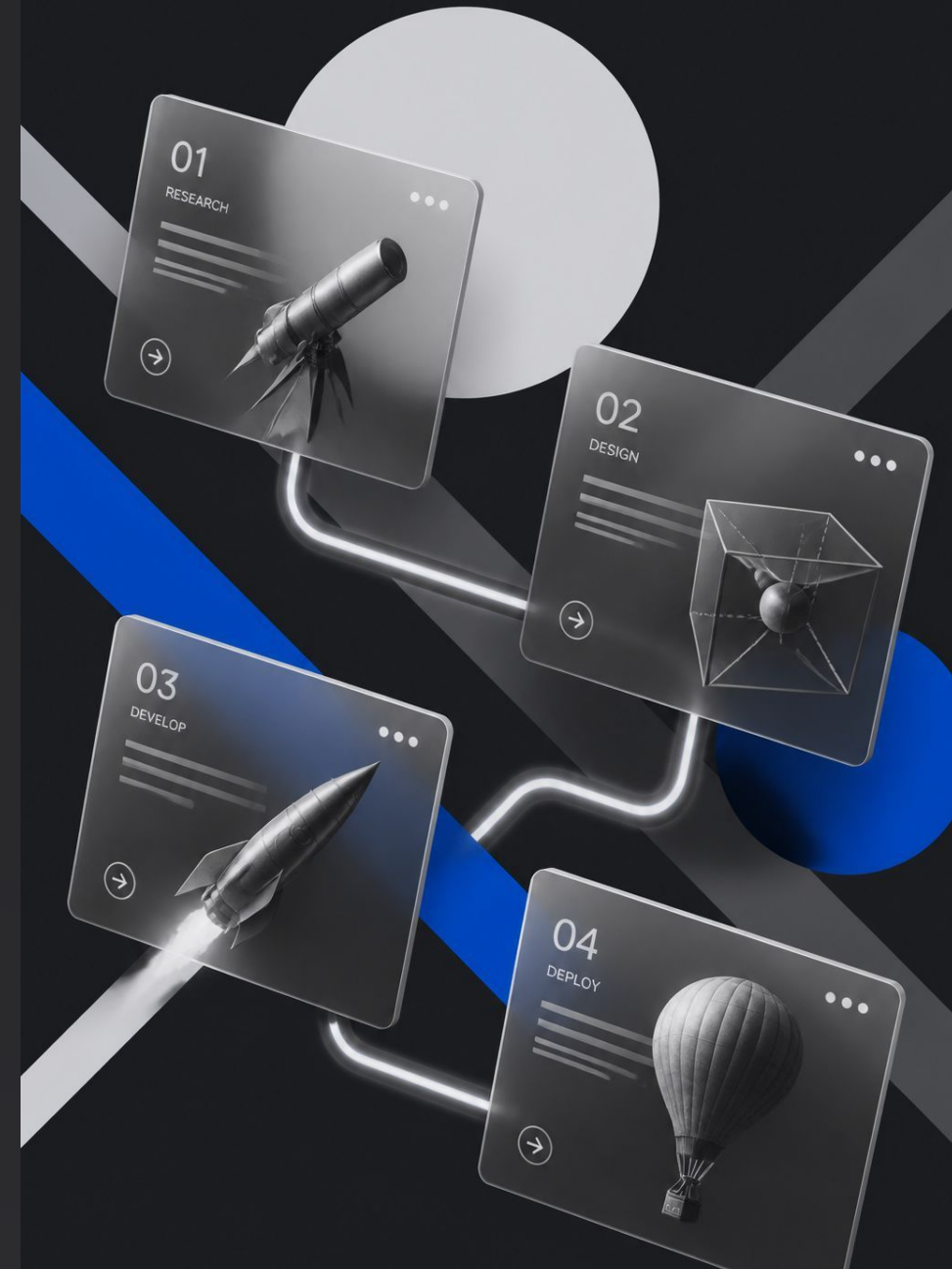
Offer Validation · Invoice Generation

System validates eligibility offline; generates a signed QR invoice instantly.

04

Invoice Sharing · Local Storage · Analytics

Invoice shared peer-to-peer; stored locally; analytics updated in real time.



Technology Stack



Frontend

React · Vite ·
Tailwind CSS ·
Zustand · Framer
Motion

Backend

Cloudflare Workers ·
Worker KV

Networking

WebRTC · PeerJS ·
ntfy.sh

Security

WebAuthn · AES-256-GCM

Database

IndexedDB · Dexie.js

Security by Design

Every layer of Spendly is engineered around a single principle: **your data belongs to you, and only you.**



Zero-Knowledge Architecture

No server ever sees your financial data. Not in transit. Not at rest.



WebAuthn + Biometric Unlock

Hardware-backed authentication via fingerprint or face — no passwords, ever.



AES-256-GCM Encryption

Military-grade encryption applied to every record stored in IndexedDB locally.



Offline-First by Default

Full functionality without internet. Sync occurs only when a trusted peer is in range.



Key Features

Expense Tracking

Real-time categorization with local analytics — no cloud required.

Merchant Billing

Instant QR invoice generation with offline inventory lookup.

Cash Wallet + EMI

Track cash flow and EMI reminders without any bank integration.

Offline Sync

Peer-to-peer synchronization resumes automatically when devices reconnect.



Advantages & Future Scope



CURRENT ADVANTAGES

Zero Server Cost

Edge-only infrastructure eliminates recurring hosting expenses entirely.

Privacy First

No telemetry, no tracking, no third-party data exposure — ever.

Fast Offline Billing

Transactions complete in milliseconds, independent of network state.

FUTURE ROADMAP

AI Analytics · Voice Billing

On-device intelligence for spending insights and hands-free merchant input.

NFC · UPI · GST · Multi-Store

Native payment rails, compliance tooling, and scalable merchant management.

Conclusion

Spendly redefines personal and merchant finance — built on zero-knowledge architecture, peer-to-peer communication, and an uncompromising commitment to privacy. No servers. No exposure. No compromise.

Privacy-First

Zero-knowledge by design

Offline-First

Works without internet

Peer-to-Peer

No central authority

Thank You

Questions & Answers

